

Gradient-intercept form



1 Consider the equation of a line $y = mx + c$.

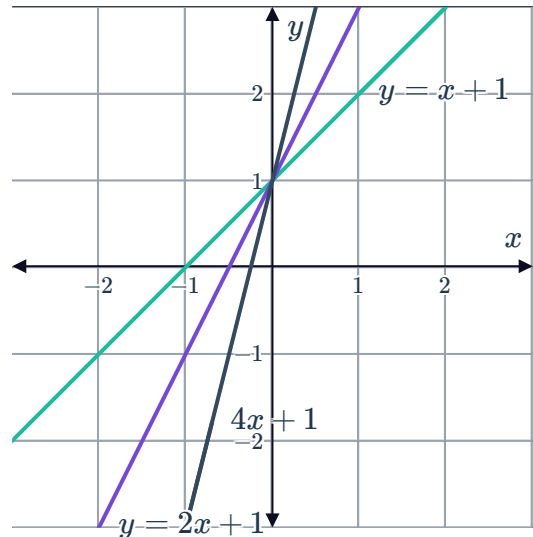
a What does m represent?

b What does c represent?

2 Consider the following three linear equations with their graphs plotted on a number plane:

- Equation 1: $y = x + 1$
- Equation 2: $y = 2x + 1$
- Equation 3: $y = 4x + 1$

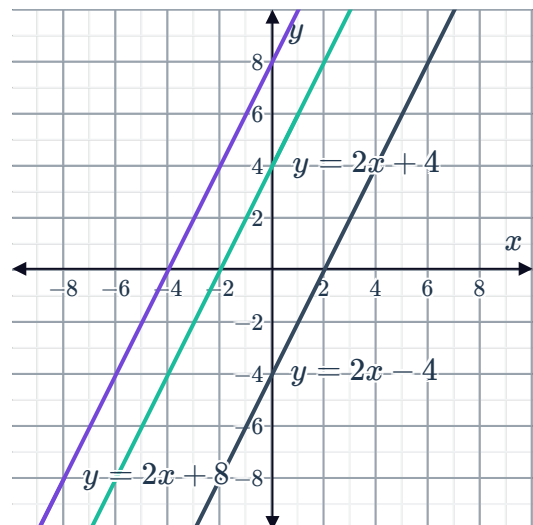
- a What do all of the equations have in common?
- b What do all of the graphs have in common?
- c Describe all lines that have the form:
 $y = mx + 1$



3 Consider the following three linear equations with their graphs plotted on a number plane:

- Equation 1: $y = 2x + 4$
- Equation 2: $y = 2x + 8$
- Equation 3: $y = 2x - 4$

- a What do all of the equations have in common?
- b What do all of the graphs have in common?
- c Describe all lines that have the form:
 $y = 2x + c$



4 Determine whether or not the gradients of the following pairs of equations are equal:

a i $y = 7x$ ii $y = 7x - 4$

b i $y = 7x + 2$ ii $y = \frac{x}{7} + 4$

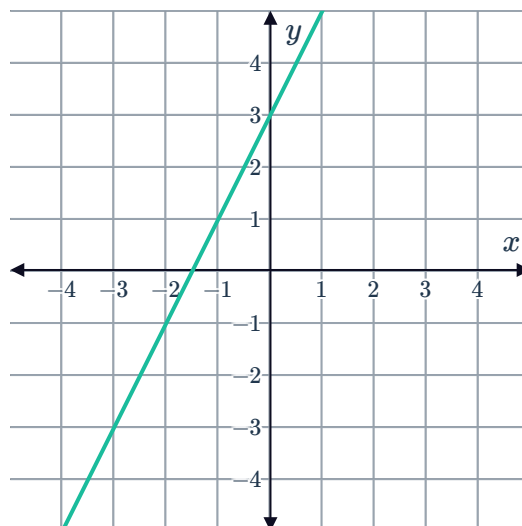
c i $y = 6 - 7x$ ii $y = 6x + 7$

d i $y = 4 + 7x$ ii $y = 7x - 10$

5 Consider the line plotted below:

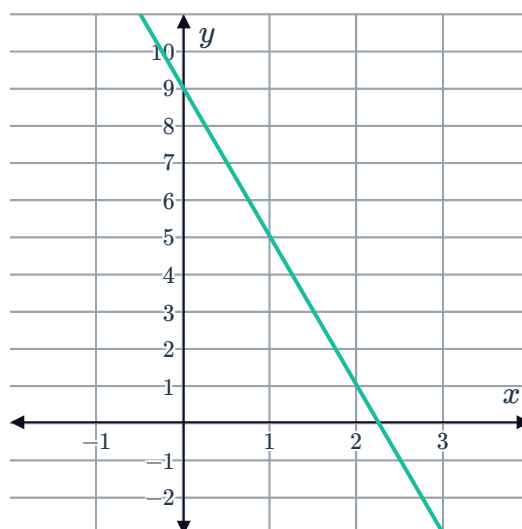


- a State the the y -intercept.
- b State the gradient.



6 Consider the line plotted below:

- a By how much does the y -value change as the x -value increases by 1?
- b State the gradient of the line.



7 State whether or not the following lines have a y -intercept:

- | | | | |
|----------------|-------------|-----------------|-----------------|
| a $y = 2$ | b $5x = 4y$ | c $y = -4x + 1$ | d $x = 1$ |
| e $y = 6x - 2$ | f $y = 3x$ | g $2x + 6y = 3$ | h $y + 4 = -5x$ |

8 Consider the following linear equations:

- i State the value of the gradient, m .
- ii State the y -intercept, c .

- | | | | |
|-------------------|-------------------|-------------------|---------------------------|
| a $y = -2x + 9$ | b $y = -4x - 8$ | c $y = 8x + 6$ | d $y = -1 - \frac{9x}{2}$ |
| e $3y = 12x - 15$ | f $-9x + 9y = 27$ | g $3x - 10y = -2$ | h $16x + 12y = 10$ |

9 Sketch the following lines on a number plane



- a The line with a y -intercept of -2 and gradient of -3 .
- b The line with a y -intercept of 3 and gradient of $-\frac{3}{2}$.
- c The line $y = 2x + 5$.
- d The line $y = \frac{1}{2}x - 1$.

10 For each of the following equations:

- i Find the y -value of the y -intercept of the line.
- ii Find the x -value of the x -intercept of the line.
- iii Find the value of y when $x = 3$.
- iv Sketch the line on a number plane.

a $y = -2x$

b $y = \frac{2x}{3} - 4$

11 For each of the following lines:

- i Find the y -coordinate of the y -intercept of the line.
- ii Hence, write the equation of the line in gradient-intercept form.
- iii Find the x -coordinate of the x -intercept of the line.
- iv Sketch the line on a number plane.

a A line has gradient $\frac{4}{5}$ and passes through the point $(-10, 4)$.

b A line has gradient -2 and passes through the point $(3, -8)$.

Equation of a line

12 A line has a gradient of -3 and intercepts the y -axis at -2 .

- a Find the equation of the line in the form $y = mx + c$.
- b State whether the point $(-2, 4)$ lies on this line.

13 A line has a gradient of -3 and cuts the y -axis at 8 .

- a Find the equation of the line in the form $y = mx + c$.
- b State whether the point $(8, -31)$ lies on this line.

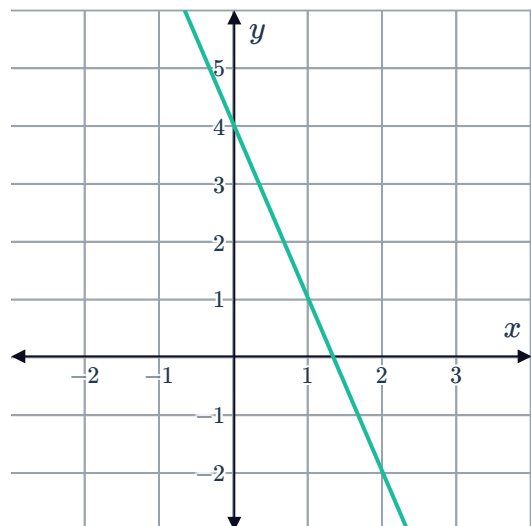
14 Find the equations of the following in the form $y = mx + c$:



- a A line that has the same gradient as $y = 9 - 8x$ and the same y -intercept as $y = -5x - 3$.
- b A line whose gradient is 2 and crosses the y -axis at 5.
- c A line whose gradient is -6 and crosses the y -axis at 9.
- d A line whose gradient is -8 and crosses the y -axis at -9 .
- e A line whose gradient is $-\frac{3}{4}$ and intercepts the y -axis at 3.
- f A line whose gradient is $\frac{4}{3}$ and goes through the point $(0, 3)$.
- g A line that has a gradient of -2 and passes through $(-6, -3)$
- h A line whose gradient is 8 and goes through the point $(0, -4)$.
- i A line whose gradient is 0 and goes through the point $\left(0, \frac{2}{13}\right)$.

15 Consider the line plotted on the number plane.

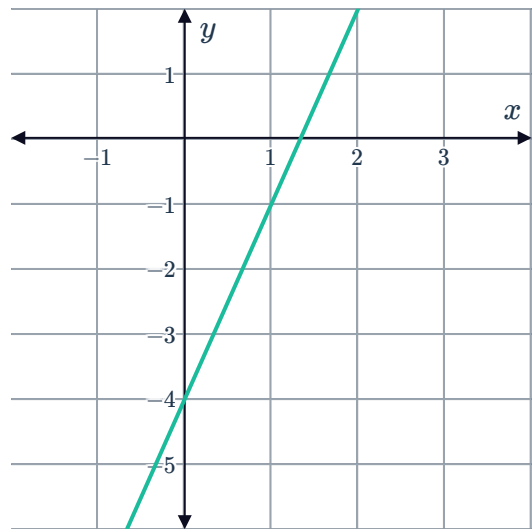
- a State the values of:
 - i The gradient, m .
 - ii The y -intercept, c .
- b Find the equation of the line in gradient-intercept form.
- c Find the value of y when $x = 27$.



16 Consider the line plotted on the number plane

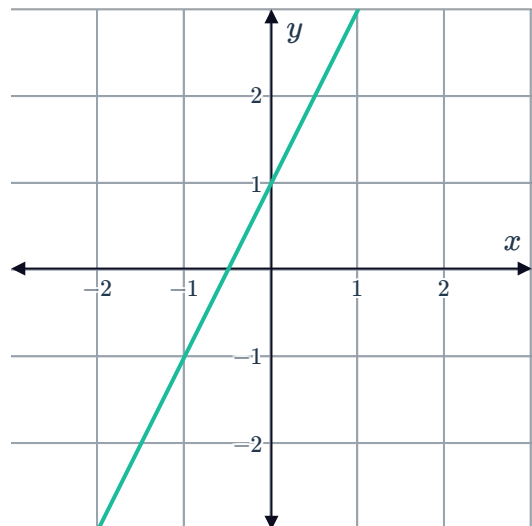


- a Find the equation of the line in gradient-intercept form.
- b Find the value of y when $x = 50$.



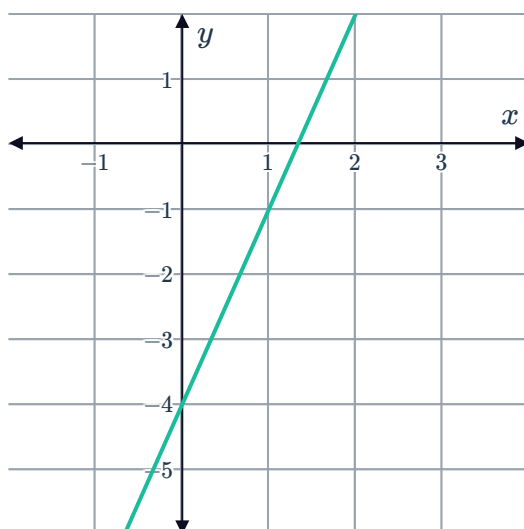
17 Consider the line plotted on the number plane.

- a Find the equation of the line in gradient-intercept form.
- b Find the value of y when $x = 29$.

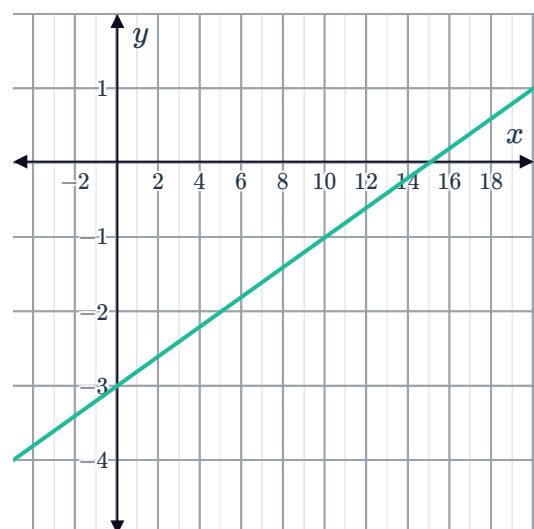


18 Find the equations of the following lines in gradient-intercept form:

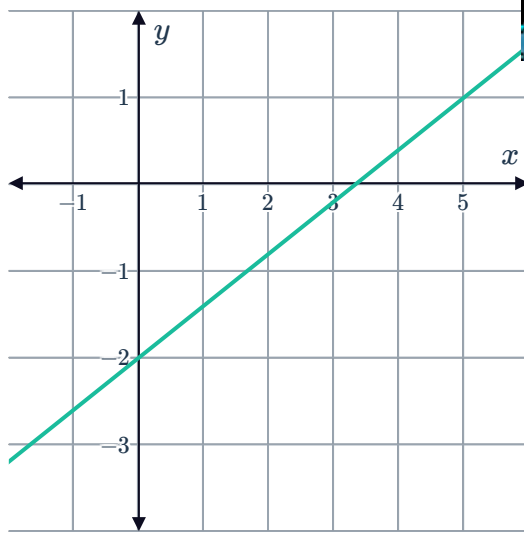
a



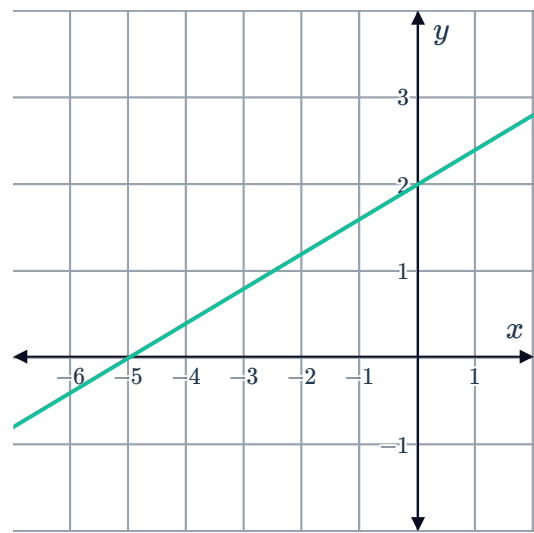
b



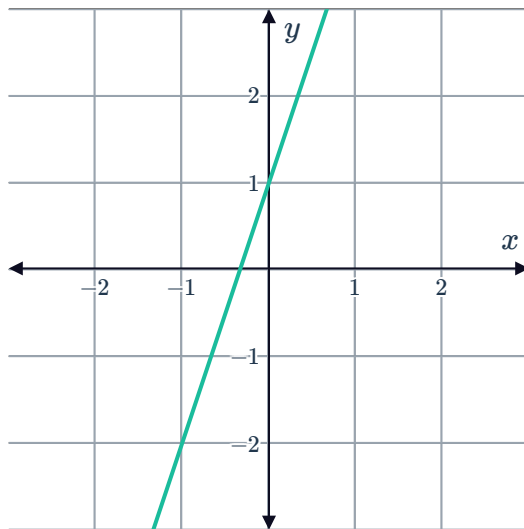
c



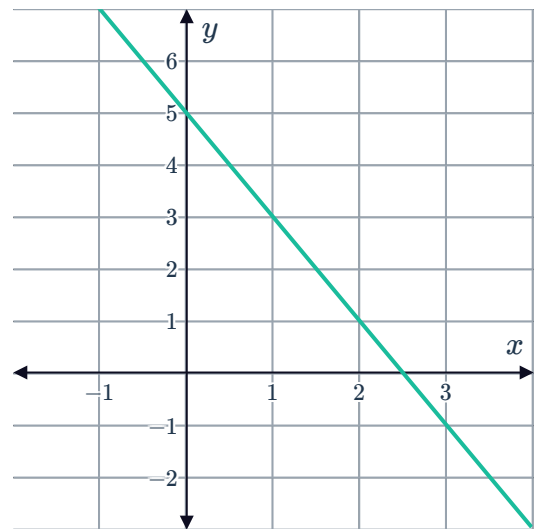
d



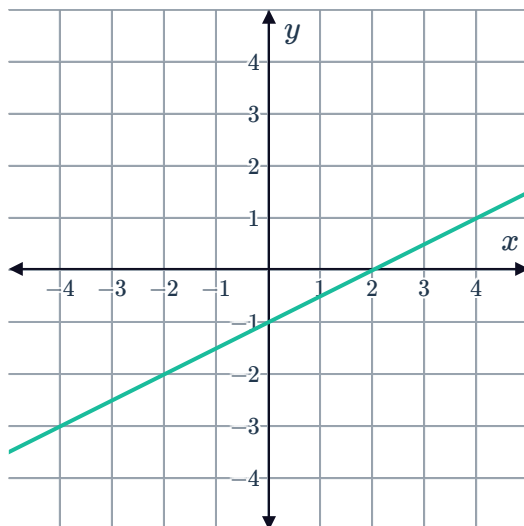
e



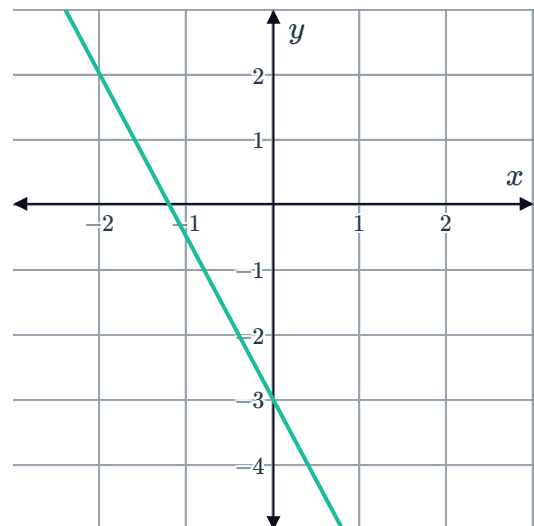
f



g



h



19 For the following lines passing through the given two points:



- i Hence, state the gradient of the line.
- ii Find the equation of the line in the form $y = mx + c$.

- a $(0, 2)$ and $(2, 6)$
- b $(0, -9)$ and $(5, 1)$
- c $(0, 2)$ and $(-7, 44)$
- d $(0, -3)$ and $(1, 4)$

20 For each of the following linear equations:

- i Rewrite it in the form $y = mx + c$.
- ii State the gradient of the line, m .
- iii State the y -intercept of the line, c .

- a $y = \frac{-4x + 16}{4}$
- b $9x - y - 8 = 0$.
- c $y = 3(4x - 3)$
- d $y = 6(3x - 2)$
- e $3x - 9y - 27 = 0$.
- f $3x - 4y - 28 = 0$

21 Consider the lines with the following equations:

- Line A: $5x + 3y + 5 = 0$
- Line B: $7x + 6y - 3 = 0$

- a Express the lines in the form $y = mx + c$.
- b State which line is steeper, A or B.

22 Determine which of the following lines are steeper: $2x + 5y - 5 = 0$ or $4x + 4y + 1 = 0$.

23 A straight line has gradient -1 and goes through the points $(0, 2)$ and $(a, -6)$.

- a Write the equation of the line in the form $y = mx + b$.
- b Find the value of a .



Gradients of parallel lines

24 Determine whether the following statements about two parallel lines are true or false.

- a** The y -value is changing at the same rate on both lines.
- b** They intersect at one point.
- c** They have the same value of c in $y = mx + c$.
- d** They have the same value of m in $y = mx + c$.
- e** They are equidistant from each other.

25 State whether the given pairs of lines are parallel:

- a** $y = -2x - 5$ and $y = -2x - 8$
- b** $y = 7x + 8$ and $y = -5x + 8$
- c** $y = -3x - 2$ and $y = -3x + 9$
- d** $y = -6x - 5$ and $y = -6x$

26 Find the gradient of the following lines:

- a** A line parallel to a line with gradient -2 .
- b** Any line that is parallel to $y = -7 + 4x$.

27 State whether the following lines are parallel to $y = 7x + 3$.

- a** $y = 7x - 3$
- b** $y = 6x + 3$
- c** $y = 7x$
- d** $y = -7x + 3$

28 State whether the following lines are parallel to $y = -3x + 2$.

- a** $y = 3x$
- b** $y = -\frac{2x}{3} + 8$
- c** $-3y - x = 5$
- d** $y = -10 - 3x$



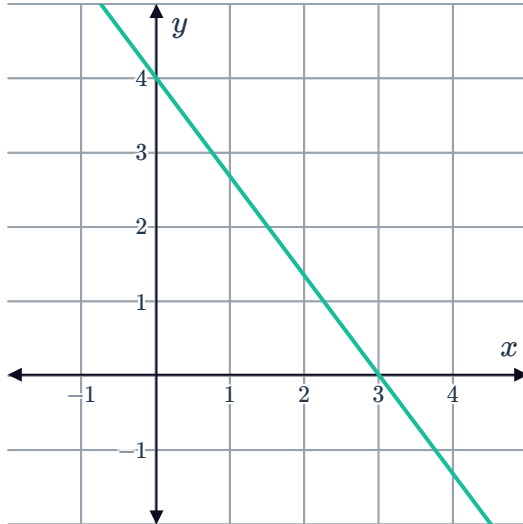
Point-gradient formula

29 For the following graphs:

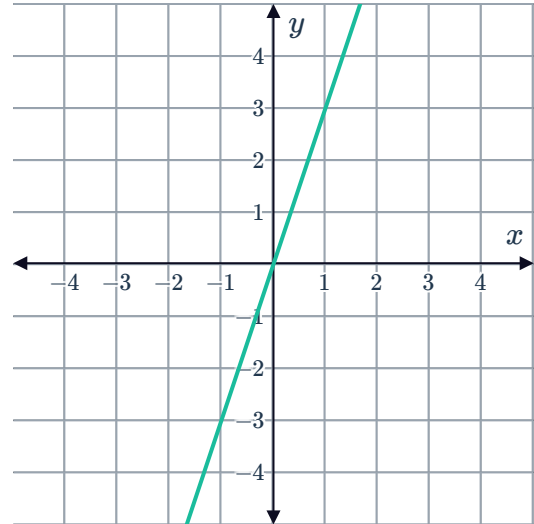
i State the value of the x -intercept.

ii State the value of the y -intercept.

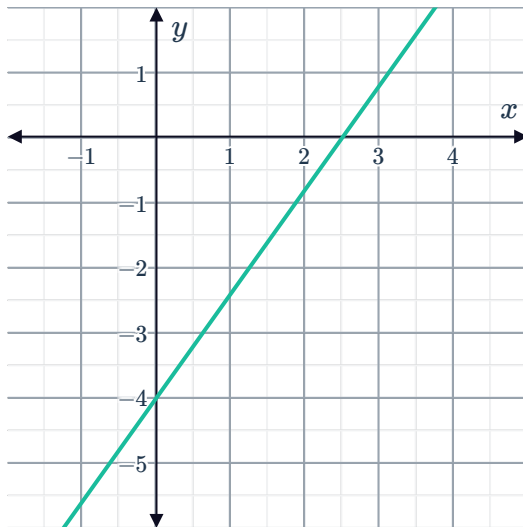
a



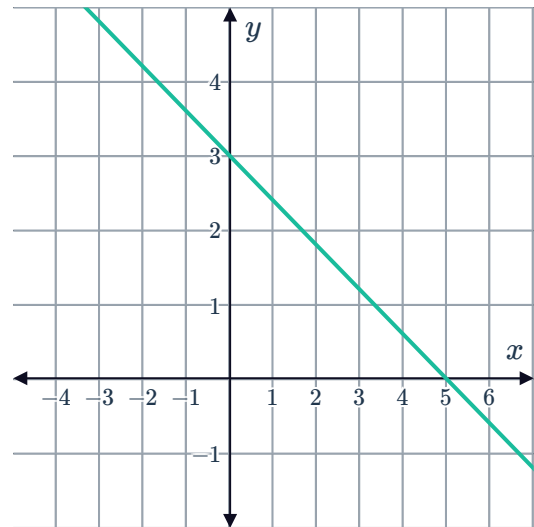
b



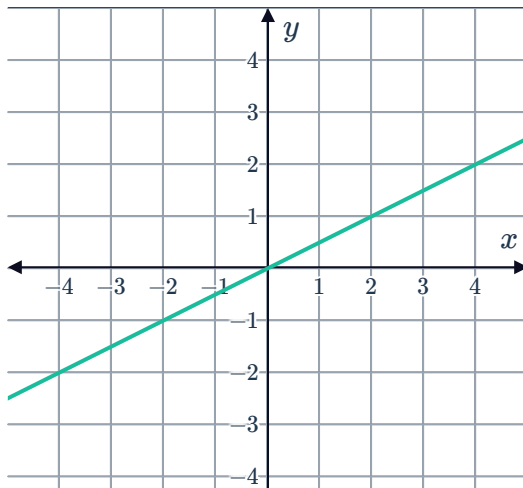
c



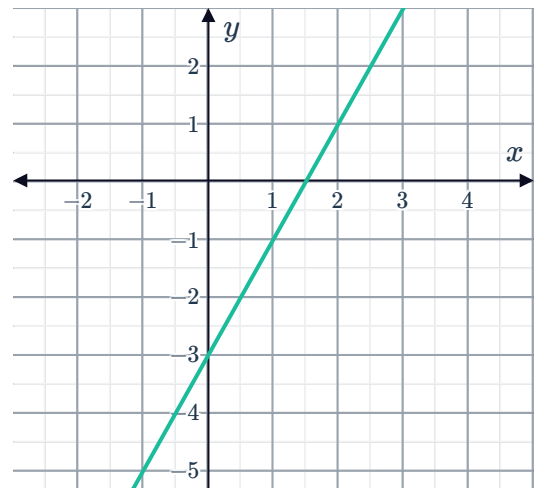
d



e



f





30 For each of the following tables of values:

- i Find the gradient, m .
- ii Find the y -intercept, c .
- iii Write the equation of the line expressing the relationship between x and y .
- iv Complete the table of values.

a

x	0	1	2	3	4	24
y	0	4	8	12	16	

b

x	0	1	2	3	4	21
y	9	14	19	24	29	

c

x	0	1	2	3	4	25
y	-23	-21	-19	-17	-15	

d

x	0	1	2	3	4	70
y	27	22	17	12	7	

31 For each of the following tables of values:



- i Find the equation of the line expressing the relationship between x and y .
- ii Complete the table of values.

a

x	1	2	3	4	19
y	5	10	15	20	

b

x	1	2	3	4	16
y	-3	-6	-9	-6	

c

x	1	2	3	4	60
y	1	4	7	10	

d

x	1	2	3	4	80
y	-14	-22	-30	-38	

e

x	1	2	3	4	-15
y	82	74	66	58	

32 Find the equation that corresponds to each of the following tables:

a

x	-1	0	1	2	3
y	7	4	1	-2	-5

b

x	1	2	3	4	5
y	5	8	11	14	17

c

x	-8	-7	-6	-5	-4
y	-37	-32	-27	-22	-17

d

x	3	4	5	6	7
y	-1	-3	-5	-7	-9

33 Find the equation of the following lines:



- a** A line that passes through the point A $(-5, -4)$ and has a gradient of -4 .
- b** A line that passes through the point A $\left(-\frac{4}{5}, -4\right)$ and has a gradient of 2 .
- c** A line that passes through Point A $(-4, 3)$ and has a gradient of 4 .
- d** A line that passes through Point A $(7, -6)$ and has a gradient of -3 .
- e** A line that passes through the point A $(3, 5)$ and has a gradient of $-\frac{5}{2}$.
- f** A line that passes through the point A $(4, 3)$ and has a gradient of $-3\frac{1}{3}$.
- g** A line that passes through the point A $(-4, 3)$ and has a gradient of -9 .
- h** A line that passes through the point A $\left(-\frac{5}{9}, 7\right)$ and has a gradient of 7 .
- i** A line that passes through the point A $(8, 1)$, and has a gradient of $\frac{5}{2}$.
- j** A line that passes through the point A $(-4, 5)$ and has a gradient of $3\frac{1}{2}$.

34 For each of the following lines:

- i** Find the equation of the line.
 - ii** Sketch the graph of the line.
- a** A line has gradient 2 and passes through the point $(-5, -3)$.
 - b** A line has gradient $-\frac{3}{2}$ and passes through the point $(-2, 2)$.
 - c** A line has gradient $-\frac{2}{5}$ and passes through the point $(-10, 2)$.
 - d** A line has gradient -3 and passes through the point $(2, -12)$.

35 Consider the line with equation $2x + y - 8 = 0$.

- a** Find the x -intercept of the line.
- b** Hence, find the equation of a line with a gradient of -4 that passes through the x -intercept of the given line.

36 Find, in general form, the equation of a line which has a gradient of $\frac{4}{7}$ and cuts the x -axis at -10 .

37 For each of the following lines:



- i Find the gradient of the line.
 - ii Find the equation of the line.
-
- a A line passes through the points $(2, -7)$ and $(-5, 6)$.
 - b A line passes through the points $(3, -3)$ and $(5, -11)$.
 - c A line passes through the points $A(-6, 7)$ and $B(-8, -4)$.

38 Identify which of the following equations of straight lines have a gradient of 5 and pass through the point $A(-1, -4)$:

- a $\frac{y+4}{x+1} = 5$
- b $\frac{x+1}{y+4} = 5$
- c $\frac{-4-y}{-1-x} = 5$
- d $\frac{y+1}{x+4} = 5$

39 Write down the equations of three lines that pass through the point $(1, 3)$. Explain how your lines are different.

Applications

40 A line has a gradient of $\frac{3}{10}$ and passes through the midpoint of $A(-6, -6)$ and $B(8, 8)$.

- a Find the coordinates of M , the midpoint of AB .
- b Find the equation of the line in general form.

41 Consider the lines L_1 , $y = -4x + 5$, and L_2 , $y = x - 1$.

- a Find the midpoint M of their y -intercepts.
- b Find the equation of the line that goes through the point M and has gradient $\frac{1}{3}$. Express the equation in general form.

42 A circle with centre $C(11, 13)$ has a diameter with end points $A(5, 14)$ and $B(p, q)$.

- a Find the value of p .
- b Find the value of q .
- c Find the equation of the line passing through B with gradient $\frac{9}{2}$.